Knowledge Assistant (RAG System)

Requirements Specification & Project Proposal • Version 1.0 • 10 November 2026

Executive Summary

New interns joining the organization often struggle to understand internal rules, policies, and procedures. This information is spread across long PDF handbooks and regulations, is hard to search, and is usually learned only by repeatedly asking supervisors and colleagues. This slows onboarding, causes avoidable mistakes, and consumes experienced staff time.

This proposal describes a web-based **Knowledge Assistant** built on Retrieval-Augmented Generation (RAG). An administrator uploads official PDF documents; interns and other users then ask questions in plain language and receive clear, **grounded answers with source citations** drawn only from the approved documents. The solution uses a Python FastAPI backend, a vector database, hybrid (semantic + keyword) retrieval, and a simple sidebar-based interface with *Admin Upload* and *Chat* views.

1. Project Overview and Goals

Problem statement. Interns find rules and policies difficult to locate and interpret, which leads to repeated questions, inconsistent answers, slow onboarding, and compliance risk.

Goals

- Give interns instant, accurate answers to rule and policy questions, available at any time.
- Ensure every answer is based on official documents and shows where it came from.
- Reduce the time supervisors and HR spend answering repetitive onboarding questions.
- Provide a single, controlled place where approved knowledge is managed by an administrator.

2. Key Users and Roles

Role	Description	Permissions
Administrator	Designated staff member (e.g., HR or program coordinator) who owns the knowledge base.	Log in; upload, view, and delete PDF documents; access Chat.
End User (Intern / Staff)	Person seeking answers about rules, policies, and procedures.	Log in; use Chat; view own conversation history. No upload access.
System Owner (Stakeholder)	Sponsor who approves scope and reviews outcomes.	Reviews reports and acceptance results.

3. Functional Requirements

ID	Requirement
FR-1	**Admin-only upload:** Only authenticated administrators can upload PDF documents; all other users cannot see or access the upload view.
FR-2	**Document management:** Admin can view the list of uploaded documents with status (processing / ready / failed) and delete a document.
FR-3	**Indexing:** Each uploaded PDF is text-extracted, split into chunks, converted to embeddings, and stored in a vector database with its metadata.
FR-4	**Hybrid retrieval:** For each question, the system combines semantic (vector) search with keyword-based search and merges the results into one ranked list.
FR-5	**Optional reranking:** A reranking step may re-order the top candidate chunks by relevance before answer generation (configurable on/off).
FR-6	**Grounded answers:** The answer is generated only from retrieved content, with source citations (document name and page). If no relevant content exists, the system states that it cannot find the answer.
FR-7	**Chat interface:** Users type questions and receive answers in a conversational chat window.
FR-8	**Conversation history:** The current session keeps recent messages so follow-up questions are understood; users can start a new chat or clear history.
FR-9	**Sidebar navigation:** A sidebar provides two views – *Admin Upload* (visible to admins only) and *Chat*.
FR-10	**Feedback (optional):** Users can mark an answer as helpful or not helpful to support later improvement.

4. Non-Functional Requirements

- **Usability:** Clean, simple interface that a new intern can use without training; responsive on desktop and tablet browsers.
- **Reliability:** Target availability of 99% during working hours; failed uploads are reported clearly and can be retried.

- **Maintainability:** Modular components (ingestion, retrieval, generation, UI) so models or databases can be replaced without redesign.
- **Scalability:** Supports growth from tens to hundreds of documents without architectural change.
- **Transparency:** Every answer is traceable to its source passages; key events are logged for auditing.

5. System Architecture Summary

A three-layer web application:

- **Frontend (web UI):** Sidebar layout with Admin Upload and Chat views; communicates with the backend over a REST API.
- **Backend (Python FastAPI):** Handles authentication, document ingestion, hybrid retrieval, optional reranking, prompt assembly, and answer generation.
- **Data layer:** Vector database for embeddings and a keyword index for lexical search; a lightweight relational store for users, document records, and chat history; file storage for original PDFs.

Query flow: User question → retrieve (semantic + keyword) → merge and optionally rerank → select top chunks → language model generates a cited answer → response shown in chat.

6. Data Handling and Document Processing

- **Accepted input:** Text-based PDF files up to an agreed size limit (e.g., 20 MB); other formats are rejected with a clear message.
- **Processing pipeline:** validate file → extract text and page numbers → clean text → split into overlapping chunks → generate embeddings → index in vector and keyword stores.
- **Metadata kept per chunk:** document title, page number, upload date, and uploader.
- **Updates and deletion:** Deleting a document removes all of its chunks from every index; re-uploading replaces the previous version.
- **Chat data:** Conversation history is stored per user and can be cleared by that user; retention period is set by the organization.

7. Security and Access Control

- **Authentication:** All users sign in; sessions use secure tokens with expiry.
- **Role-based access:** Upload and delete functions are enforced as admin-only on the server, not just hidden in the UI.
- **Data protection:** Encrypted connections (HTTPS); secrets and API keys kept outside source code; uploaded files validated and scanned for type and size.
- **Privacy:** Users can only see their own conversations; document content is not shared outside the approved services.
- **Auditability:** Uploads, deletions, and login events are logged.

8. Performance Expectations

Measure	Target
Answer response time	Typically within 5 seconds for 90% of questions (first words streamed sooner).
Document processing	A 50-page PDF is searchable within about 2 minutes of upload.
Concurrent users	At least 25 simultaneous chat users without noticeable slowdown.
Knowledge base size	Up to about 200 documents / 20,000 chunks in the initial release.

9. Acceptance Criteria

- A non-admin user cannot upload, delete, or open the Admin Upload view, even by calling the API directly.
- An uploaded PDF becomes searchable and an answer to a question from its content cites the correct document and page.
- Questions with no supporting content receive a clear “not found in the documents” response rather than a guess.
- Follow-up questions (e.g., “and what about interns?”) are answered using earlier messages in the same chat.
- In a pilot test with at least 30 representative questions, 85% or more of answers are rated correct and grounded by reviewers.

10. Scope Limitations

In scope: PDF ingestion, hybrid retrieval, grounded chat, simple history, admin/user roles. **Out of scope (initial release):** OCR for scanned PDFs, non-PDF formats, voice input, multi-language support, mobile apps, integration with external HR or learning systems, and per-document permissions.

11. Assumptions and Risks

Assumptions	Risks and mitigation
• Source documents are official, current, and text-based PDFs. • An administrator is assigned to keep documents up to date. • Approved access to a language model and embedding service is available. • Users have a modern web browser and network access.	• **Inaccurate answers** – mitigated by grounding, citations, and "not found" behavior. • **Outdated content** – mitigated by admin review and re-upload process. • **Poor PDF text quality** – mitigated by upload validation and warnings. • **Sensitive data exposure** – mitigated by role-based access and logging.

Final Acceptance Checklist

- ☐ Admin-only upload enforced on both UI and server
- ☐ PDFs processed, chunked, embedded, and stored in the vector database
- ☐ Hybrid retrieval (semantic + keyword) working; reranking configurable
- ☐ Answers are grounded and cite document and page
- ☐ "Not found" response shown when no relevant content exists
- ☐ Chat interface with sidebar (Admin Upload and Chat views) delivered
- ☐ Conversation history supports follow-up questions
- ☐ Performance targets met in pilot testing
- ☐ Pilot accuracy of 85% or higher confirmed by reviewers; stakeholder sign-off obtained

Approved by: _____ Role: _____ Date: _____